NZK Heating System (MACRO) - Assumptions Report

Model: heating_real v2 | Year: 2021 | 17 nodes | 212 assets (17 types) | Units: MW, USD/MW, USD/MW-yr, USD/MWh

1. Structure

• Per node: gas node, electricity node, DH heat node *heat_dh* (11 nodes with DH), decentral heat node *heat_dec* (all 17), industrial steam node *steam_ind* (12 nodes), waste node (DH nodes). CO2 goes to one national node *co2_KOR*.
• No heat network between nodes; gas and CO2 have no inter-node network. Temperature enters only through the demand series and the ASHP COP series.
• DH nodes: those with ≥ 200,000 Gcal of district heat in 2021 (SEL, PUS, TAE, INC, KWJ, SJG, GGI, CNA, CNB, DJJ, GNA). Non-DH nodes choose between gas boiler, ASHP and electric heater.
• Cooling is not modelled (Ninja cooling_demand downloaded but unused).

2. Demand

Item	Assumption / source
Hourly shape	Renewables.ninja Demand.ninja (MERRA-2, 2021), *heating_demand*; mean of ~20 km grid cells (271 cells) per province, weighted by land fraction; shifted UTC to KST (+9 h) and normalised to sum 1.
DH demand (heat_dh)	District heating (jiyeok nanbang) TOE by province 2021 x 10 Gcal/TOE x 1.163 MW per Gcal/h. Source: Dropbox heat_demand.xlsx.
Decentral demand (heat_dec)	City-gas TOE by province 2021 x 10 Gcal/TOE x 0.7 (ASSUMED share of gas used for space heating + DHW) x 1.163.
Industrial steam (steam_ind)	Industrial heat demand by province 2021 (Korea Energy Agency, Gcal), flat profile, weekend -5% (normalised). Nodes with ≥ 300,000 Gcal.
Known limitations	Same Ninja shape for DH and DEC; no DHW base load (summer demand ~0; KDHC shows 7-16% of annual in Jun-Aug); peak/mean ~6 vs ~3 in KDHC data, which inflates peak capacity.

3. Global parameters

Parameter	Value	Note
Discount rate (WACC)	10%	all assets
Natural gas price	35 USD/MWh (LHV)	ASSUMED, national, same in all nodes
Gas emission factor	0.202 tCO2/MWh fuel	CHP, boilers, industrial CHP
MSW gate fee	-15 USD/MWh waste	negative variable cost
WtE emission factor	0.35 tCO2/MWh waste, 50% biogenic	ASSUMED
Unit conversions	1 TOE = 10 Gcal; 1 Gcal/h = 1.163 MW	

4. Asset parameters (same values at all nodes of the type unless noted)

Inv = investment cost (USD/MW); FOM = fixed O&M (USD/MW-yr); VOM = variable O&M (USD/MWh); Life = lifetime = capital recovery period (yr). All cost values are generic ASSUMPTIONS to be replaced with Korean data.

Asset type	Nodes	Technical parameters	Inv	FOM	VOM	Life	Expand / Retire	Capacity basis	Existing / Max capacity
FuelSupply	17	gas, no capacity limit	-	-	35	-	-	-	unlimited
WasteSupply	11 DH	MSW	-	-	-15	-	No / -	MW waste LHV	existing = WtE input; max = max(3x existing, 50)
CHP_gas	11 DH	eta_elec_max 0.55; eta_total 0.85; cv 0.15; cb 0.90; min load 30%	1,200,000	36,000	3.5	30	Yes / Yes	MW elec (condensing)	existing from KDHC 2021 elec (ccgt+CES+fuel cell)/8760/CF 0.60 (SEL 14, GGI 2,252, others 0); max = max(peak DH, 1.5x existing)
CHP_waste (WtE)	11 DH	eta_elec_max 0.25; eta_total 0.80; cv 0.10; cb 0.30; min load 50%	3,500,000	120,000	8	30	Yes / No	MW waste LHV in	existing from KDHC 2021 elec (waste+biomass+biogas)/8760/CF 0.85/0.25 (SEL 56, TAE 71, CNB 40, GGI 22, KWJ 9); max = 3x existing (min 50)
HeatOnlyBoiler	11 DH	efficiency 0.92	90,000	1,800	1	25	Yes / Yes	MW heat out	existing 0.9 x DH peak (ESTIMATE); max 1.5 x peak
ElectricBoiler	11 DH	efficiency 0.99	70,000	1,000	0.5	20	Yes / No	MW heat out	existing 0; max 0.5 x DH peak
LargeHeatPump	11 DH	COP 2.8 (fixed)	900,000	8,000	2	20	Yes / No	MW heat out	existing 0; max 0.5 x DH peak
ThermalStorage	11 DH	charge/discharge eff 0.98; loss 0.3%/h; duration 2-24 h; symmetric charge/discharge	15,000 (power) + 3,500 (USD/MWh energy)	300 (power) + 35 (energy)	0	30	Yes / No	MW discharge	existing 0.08 x DH peak power, 6 h; max 0.5 x peak power, 24 h
IndustrialWasteHeat	INC	availability 0.85; steel/petrochem	250,000	3,000	3	30	Yes / No	MW heat out	existing 0; max 120
WasteHeatHP	SEL, GGI	COP 4.2; availability 0.90; data centres (SEL), fabs (GGI)	600,000	6,000	2	20	Yes / No	MW heat out	existing 0; max 80 (SEL), 250 (GGI)
IndustrialSteamDemand	12	flat steam demand	-	-	-	-	-	-	-
IndustrialCHP	12	back-pressure; eta_total 0.88; cb 0.60	1,000,000	30,000	3	30	Yes / Yes	MW elec	existing = 0.85 x 0.6 x steam peak (ESTIMATE); max = steam peak
SteamBoiler	12	efficiency 0.90	80,000	1,500	1	25	Yes / Yes	MW steam out	existing 1.1 x steam peak; max 1.5 x steam peak (ESTIMATE)
DecentralGasBoiler	17	efficiency 0.93	150,000	3,000	1	20	Yes / Yes	MW heat out	existing 1.05 x DEC peak (ESTIMATE); max 1.5 x peak
DecentralASHP	17	COP series f(outdoor T): clip(3.4 + 0.07(T-7), 1.8, 4.6), synthetic T	1,000,000	20,000	0	15	Yes / No	MW heat out	existing 0.03 x DEC peak; max 1.5 x peak
DecentralElectricHeater	17	efficiency 1.0	100,000	1,000	0	20	Yes / No	MW heat out	existing 0.05 x DEC peak; max 1.5 x peak
Demand (heat_dh / heat_dec)	11 / 17	hourly series from Section 2	-	-	-	-	-	MW	-

5. CHP feasibility region (P-Q)

• Extraction line: P ≤ eta_elec_max x F - cv x Q. Back-pressure: P ≥ cb x Q. Total fuel: P + Q ≤ eta_total x F. Minimum load: F ≥ min_load x F_max (with commitment variable).
• Gas CHP vertices (per unit of fuel): back-pressure point P = 0.40, Q = 0.45; maximum extraction P = 0.48, Q = 0.37. WtE: back-pressure P = 0.18, maximum extraction P = 0.19.
• The CSV carries the parameters; the constraint itself needs a custom constraint in MacroEnergy.jl (not implemented in the CSV).

6. Existing capacity by node (MW)

Node	CHP gas (elec)	WtE (waste in)	DH boiler	TES (power)	Ind. CHP (elec)	Steam boiler	Dec. gas boiler
SEL	14	56	4,346	386			23,493
PUS	0	0	324	29	23	50	7,853
TAE	0	71	689	61	46	100	4,692
INC	0	0	1,716	153	36	77	5,413
KWJ	0	9	242	22			3,093
USN					737	1,589	3,257
SJG	0	0	568	51			262
GGI	2,252	22	11,050	982	225	486	20,526
GWN							1,999
CNA	0	0	276	25	354	764	3,401
CNB	0	40	455	40	94	203	2,252
DJJ	0	0	478	42	24	52	3,085
JNB					213	458	2,992
JNA					573	1,236	2,284
GNB					158	341	4,000
GNA	0	0	508	45	55	117	5,034
JEJ							507

7. Annual demand 2021 by node (GWh)

Node	Province	DH heat	Decentral heat (70% gas)	Industrial steam	Hourly shape
SEL	Seoul	6,356	29,452	0	Ninja
PUS	Busan	344	7,155	393	Ninja
TAE	Daegu	980	5,716	779	Ninja
INC	Incheon	2,494	6,742	598	Ninja
KWJ	Gwangju	346	3,786	2	Ninja
USN	Ulsan	0	2,994	12,473	Ninja
SJG	Sejong	816	322	54	Ninja
GGI	Gyeonggi-do	16,640	26,492	3,810	Ninja
GWN	Gangwon-do	0	2,702	0	Ninja
CNA	Chungcheongnam-do	394	4,158	5,997	Ninja

Node	Province	DH heat	Decentral heat (70% gas)	Industrial steam	Hourly shape
CNB	Chungcheongbuk-do	682	2,889	1,592	Ninja
DJJ	Daejeon	692	3,827	413	Ninja
JNB	Jeollabuk-do	50	3,722	3,594	Ninja
JNA	Jeollanam-do	132	2,687	9,701	Ninja
GNB	Gyeongsangbuk-do	113	4,792	2,675	Ninja
GNA	Gyeongsangnam-do	673	5,715	916	Ninja
JEJ	Jeju	0	337	21	Ninja

8. Items to verify (priority)

#	Item	Current value	Why it matters
1	Share of city gas used for heating	70%	sets total decentral demand
2	Hourly demand shape	Ninja only; peak/mean ~6; no DHW base load	oversizes peak capacity, no summer load for CHP
3	Capacity factors for existing CHP / WtE	0.60 / 0.85; WtE eta 0.25	converts KDHC generation to MW
4	Existing CHP outside KDHC	0 in nodes without KDHC data	Incheon, Busan, Seoul etc. have other operators (GS Power, SK E&S, Seoul Energy Corp.)
5	CHP P-Q parameters	Section 4	define the feasibility region
6	Existing boilers, TES, industrial CHP	fractions of peak	estimates, no data
7	Costs, lifetimes, WACC	generic	no Korean source yet
8	Gas price, gate fee, emission factors	35 USD/MWh; -15; 0.202 / 0.35	scenario inputs
9	ASHP COP	synthetic temperature	replace with MERRA-2/ERA5 temperature from the same year
10	Node code mapping	CNA/CNB, JNA/JNB, GNA/GNB	confirm vs model location list
11	DH node threshold	≥ 200,000 Gcal	decides which nodes have DH assets
12	Cooling	not modelled	needs scale source and double-counting check vs electricity demand